

New vortex on Neptune.

NASA's hubble telescopes discovered a new dark vortex on the surface of Neptune. the first on the planet
<http://dgit.in/vortexx1>

**Measure the universe?**

Physicists create a quantum ruler to measure the universe.
<http://dgit.in/q rulerx>

Artificial meat

Lab grown vegan alternatives

Ashwin Krishnan

feedback@digit.in

Since time immemorial, the word tofu has been met with scorns of sarcasm and looks of disapproval from regular meat eaters, who consider the substitute a poor alternative to the taste and texture of actual animal in their mouths. In a lot of ways, communities and individuals that embrace vegetarianism have always been excluded from a vast range of culinary creations which hinge on the usage of meat. For some reason, taste enhancers never quite produced the same effect of popping a whole chicken nugget and savouring the flavour that explodes in the meat eater's mouth as he bites into it. This, however, is a time of modern science, and more importantly a time of environmental consciousness. With some studies showing that giving up raising cattle for beef production can have a bigger impact on reducing our carbon footprint than giving up petrol based transportation, it is vital that we start looking for solutions. Armed with the latest in genetic science, artificial meat substitutes have sprung up in super-labs all over the world. In this article we analyse how they work and see how the staunch Brahmin can soon enjoy a steaming hot plate of mutton biriyani

What makes meat meat?

To many people, the answer would probably be taste. However, scientific experiments and taste tests from the world's leading culinary experts show that capturing the taste alone is not sufficient. It's equally important to capture the experience of biting into a piece of meat and feeling the texture as it settles on your tongue. What differentiates a hamburger from a chicken patty is not only the taste,

it depends extensively on the way the patty disintegrates.

The challenge then is to painlessly, and without harmfully impacting the environment, produce meat with the same fibrous texture in a lab. This also has varying levels of difficulty. There are some food items that have very little complex structure such as processed foods. Sausages, for example, can be grown by extracting cells or serum from cows and growing it in

that are used purely to feed animals; valuable pastures that could be used to grow grain for human consumption. Almost 1,800 gallons of fresh water for drinking, feed irrigation and processing go into one pound of beef, according to National Geographic. Lab-grown food, on the other hand, requires far less land and water, and emits less greenhouse gas compared to raising livestock, according to a study by the University of Oxford. Cows and



Difference in texture: Chicken(left) and Beef(right)

the lab. Of course, this in itself is a tedious and complicated process that involves a lot of biotechnological wizardry but it is still something that can be done reasonably well. A steak, however, is made of muscle tissue which is threaded through with extremely long, fine capillaries which transport blood and nutrients directly to the cells. It is much more difficult to reproduce such a complex structure as the process involves not only cultivating cells but growing actual muscle fibre with the appropriate texture and flavour in the lab.

Why do we need artificial meat?

Apart from guilt free burger binging, constructing artificial meat has a hugely positive impact on the environment. There is a pressing need to reduce the environmental impact of rearing livestock; nearly 30% of the Earth's surface is covered with pastures

other livestock animals produce enough methane to account for a staggering 18% of total global greenhouse gas emissions.

Above and beyond all these practicalities, it is also a scientific challenge that biohackers across the world are enthused by. Creating an entirely natural structure from scratch in a laboratory is a difficult task but the intellectual obstacles involved are the kinds that scientists love to tackle.

The science of meat making

Apart from a lot of money, a couple of degrees, a doctorate and a superlab, these are the components that you will need to grow your own artificial meat substitute

- **Starter cells:** There are two extremes in terms of choosing starter cells. Stem cells, the magic words in medical research at the moment, and differentiated muscle fibres. Stem cells are cells that haven't decided what they want